

In-Service Inter-Segment Skew Calibration Using a Built-in Low-Speed Coherent Detection Monitor

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Abstract We propose a power-efficient in-service skew calibration method for the transmitters with segmented modulators, utilizing a modulated local oscillator and low-speed coherent detection monitor. Experimental results demonstrate an accurate skew calibration with a maximum error within ± 0.025 UI over a ± 0.2 UI skew range. ©2026 The Author(s)

Introduction

Driven by the escalating demands for high throughput, low power, and low latency in DCIs, researchers are actively exploring more innovative and flexible optical transceiver architectures. The transmitter architecture employing segmented modulator has emerged as a promising approach, offering notable advantages in performance and power efficiency [1-15]. This architecture divides the multi-level electrical signal into multiple binary-level signals, which are then individually modulated and synthesized into the analogue optical signal directly in the optical domain. This approach relaxes the stringent requirements for digital-to-analogue and electrical-to-optical mapping. For instance, optical digital-to-analogue converter (DAC) transmitters referenced in [14, 15] eliminate the power-intensive electrical DACs and linear drivers. Instead, they generate the analogue optical signal using a segmented modulator driven by binary gate outputs and low-power binary CMOS drivers, reducing power consumption.

However, such segmented architecture introduces a critical challenge: inter-segment timing skew. Variations in path lengths, device characteristics, and environmental factors cause timing differences among those signals [16, 17]. In the optical DAC architecture, different segments may be modulated by different bits so that the inter-segment skews significantly induce nonlinear distortions, as demonstrated in our previous behavior model [18, 19]. Consequently, accurate in-service skew calibration is needed to ensure the stable, efficient, and optimal performance of the

transmitters with segmented modulators.

In this paper, we propose a novel in-service skew calibration approach for the transmitters with segmented modulator, utilizing modulated local oscillator (MLO) and low-speed coherent detection as a built-in monitor. This in-service calibration works during normal data transmission without dedicated training symbols or interrupting traffic. The experiments show that this method is effective, achieving precise skew calibration with a maximum error of ± 0.025 UI across a ± 0.2 UI skew range.

Proposed In-Service Skew Calibration: Architecture, Principles, and Characteristics

Fig. 1(a) illustrates the proposed in-service skew calibration architecture, featuring a MLO branch and a coherent detector. This detector mixes the tapped transmitter's output with the MLO signal to extract their product. By designing the sequence $B[n]$ of MLO branch, specific desired features relevant for skew calibration can be selectively captured from this product. Therefore, the design of $B[n]$ is crucial, as it directly determines the specific bit under test (BUT) and the feature (e.g., phase or timing error) to be detected.

Taking a PAM-4 transmitter as an example, the modulator uses one equal-length segment for the least significant bit (LSB) and other two for the most significant bit (MSB), represented by binary sequences $A_{LSB}[n]$ and $A_{MSB}[n]$, respectively. Consequently, the optical DAC transmitter's optical output $E_{out}(t)$ is expressed as Eq. (1) and the differential phase $\varphi_O(t)$ is in Eq. (2).

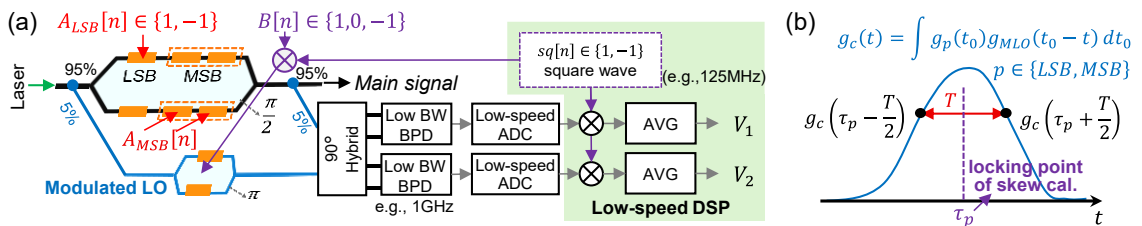


Fig. 1: (a) Schematic of proposed in-service skew calibration architecture. LSB: least significant bit. MSB: most significant bit. LO: local oscillator. BPD: balanced photodiode. AVG: averaging. (b) The underlying principle of skew monitoring.

$$E_{Out}(t) = E_0 \sin\left(\frac{\varphi_O(t)}{2} + \frac{\pi}{4}\right) \quad (1)$$

$$\varphi_O(t) = \sum_n \left(A_{LSB}[n]g_{LSB}(t - \tau_{LSB} - nT) + 2A_{MSB}[n]g_{MSB}(t - \tau_{MSB} - nT) \right) \quad (2)$$

Here, g_p (where $p \in \{LSB, MSB\}$) is the bit's impulse response, T is the symbol period, and τ_p signifies bit's time delay relative to MLO signal.

The MLO branch uses a tapped laser and a modulator biased at its null point. Its input symbol sequence is formed by multiplying the designed detection sequence $B[n]$ with a low-frequency square wave $sq[n]$ (e.g., 125 MHz). This process of square wave multiplication and corresponding multiplication in the final low-speed digital signal processing (DSP) block helps address challenges caused by unwanted DC distortions, such as modulator bias drift and PD dark current. For clarity, the derivations omit the square wave aspect. The MLO's output is given by Eq. (3),

$$E_{MLO}(t) = \alpha E_0 \sin\left(\frac{\varphi_{MLO}(t)}{2}\right) = \alpha E_0 \sin\left(\frac{1}{2} \sum_n B[n]g_{MLO}(t - nT)\right) \quad (3)$$

where α is the tapped laser ratio, and g_{MLO} is the impulse response of MLO.

Considering a phase difference θ between the tapped transmitter's output signal and MLO signal, the indicators obtained from low-speed coherent detection (involving low-bandwidth balanced PDs (BPDs), DC blocks, square wave multiplication, and averaging) are expressed as:

$$V_1 = \langle \Re(\alpha E_{Out}(t)E_{MLO}(t)e^{-j\theta}) \rangle = V_0 \cos \theta \quad (4)$$

$$V_2 = \langle \Im(\alpha E_{Out}(t)E_{MLO}(t)e^{-j\theta}) \rangle = -V_0 \sin \theta \quad (5)$$

Here, V_0 contains the desired information. Polarization and optical delay are assumed to be aligned between $E_{Out}(t)$ and $E_{MLO}(t)$ due to integrated structure. If MLO sequence $B[n]$ is only correlated with $A_p[n]$, V_0 can be further derived.

$$V_0 = \langle \alpha E_{Out}(t)E_{MLO}(t) \rangle \propto \langle \varphi_O(t)\varphi_{MLO}(t) \rangle = \frac{1}{N} \sum_{m=1}^N B[m] \sum_{n=1}^N A_p[n]g_c\left((m-n)T - \tau_p\right) \quad (6)$$

Here, $g_c(t) = \int g_p(t_0)g_{MLO}(t_0 - t)dt_0$ is the cross correlation between the impulse response of

corresponding bit and that of MLO, as shown in Fig. 1(b). When $B[n]$ is designed as $A_p[n-1] + A_p[n]$ with a T/2 clock shift, Eq. (6) simplifies to,

$$V_0 \propto R_{A,p}(0) \left(g_c\left(\frac{T}{2} - \tau_p\right) + g_c\left(-\frac{T}{2} - \tau_p\right) \right) > 0 \quad (7)$$

Here, we omit detailed derivations and use the $R_{A,p}(k) = \langle A_p[n]A_p[n+k] \rangle$ where $R_{A,p}(k) = 1$ if $k = 0$ and 0 otherwise. In this case, the V_0 in Eq. (7) is positive, thus the phase difference θ in Eq.(4-5) can be estimated as $\hat{\theta} = \arg(V_1 - jV_2)$.

To monitor skew τ_p , the MLO sequence is set to $B[n] = A_p[n-1] - A_p[n]$ with T/2 clock shift. In this case, V_0^p in Eq. (6) is proportional to τ_p ,

$$V_0^p \propto R_{A,p}(0) \left(g_c\left(\frac{T}{2} - \tau_p\right) - g_c\left(-\frac{T}{2} - \tau_p\right) \right) \approx k\tau_p \quad (8)$$

where k is the sensitivity of skew monitoring.

Tab. 1 outlines the skew calibration method. Phase estimation (step 1) and skew estimation (step 2) should be performed sequentially for each bit. Calibration involves adjusting τ_p until $V_0^p = 0$, which implies $g_c\left(\frac{T}{2} - \tau_p\right) = g_c\left(-\frac{T}{2} - \tau_p\right)$. This criterion aligns the skew between the BUT and the MLO with the midline of T -width waistline of g_c , as shown in Fig. 1(b).

Based on the above algorithms, beyond the inherent power benefits of low-speed coherent detection and simple DSP, the cost and power consumption of MLO can also be maintained to an acceptably low level through several design considerations. Firstly, as skew variations occur slowly, the proposed method could use long-time averaging to suppress noise. Thus, a small tap ratio and a small modulation swing in MLO which significantly relaxes the MLO design and power consumption are acceptable. Secondly, for the same reason of slow skew dynamics, intermittent disabling of the skew calibration module to reduce the power is also feasible. Thirdly, the MLO symbol sequence $B[n]sq[n]$ utilize only three levels $\{-1,0,1\}$, which means that only a single segment is needed for MLO modulator.

Thus, the proposed skew calibration offers:

- **Background Calibration:** Operates without disturbing transmitter's output signal.
- **Low Power Consumption:** Low-speed coherent detection and low-speed DSP allows for the use of ultra-low bandwidth components, such as 1 GHz BPDs and ultra-low-sampling-rate ADCs. The power consumption of MLO also can be limited to an acceptable level.
- **Versatile Applicability:** Capability to detect the phase differences θ extends its use to coherent transmitters, as shown in Fig. 2.

Experimental Validation and Results

As depicted in Fig. 3(a), experiments were conducted on a 32 Gbaud PAM-4 optical DAC

Tab. 1: Proposed skew calibration method: MLO signal design and step-by-step calculation formulas

	MLO signal	Cal. formula
Step 1: phase est.	$A_p[n-1] + A_p[n]$	$\hat{\theta} = \arg(V_1 - jV_2)$
Step 2: skew ind. est.	$A_p[n-1] - A_p[n]$	$V_0^p = V_1 \cos \hat{\theta} - V_2 \sin \hat{\theta}$

$p \in \{LSB, MSB\}$ Repeat steps 1 and 2 on each bit.

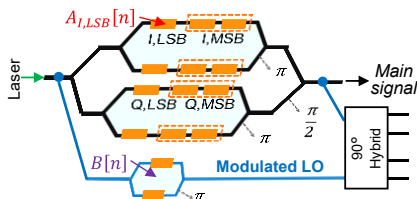


Fig. 2: Proposed optical circuit for in-service skew calibration scheme adaptable to segmented coherent transmitters.

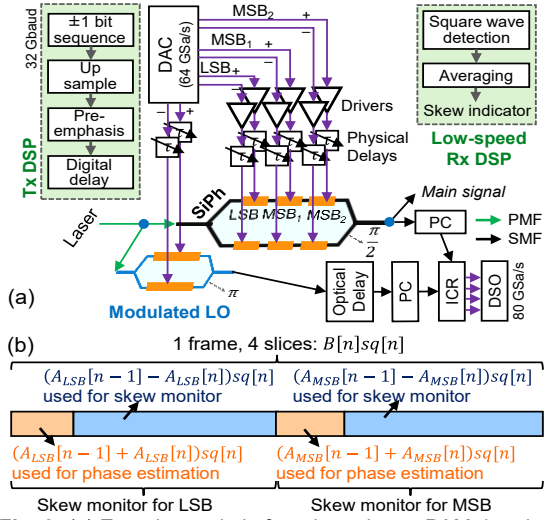


Fig. 3: (a) Experimental platform based on a PAM-4 optical DAC transmitter. (b) Frame design for MLO branch.

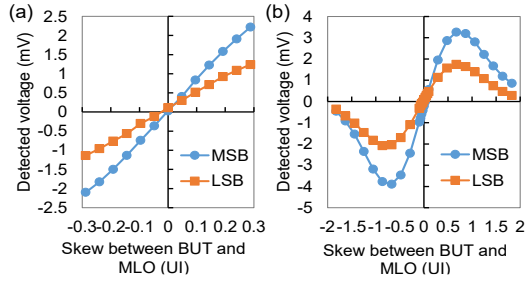


Fig. 4: (a) Detected voltage versus the skew between BUT and MLO. (b) The "S" curves of LSB and MSB.

transmitter. This transmitter employs an all-silicon modulator featuring three equal-length segments. Three differential signals of the electrical DAC drove one LSB segment and two MSB segments via drivers and precisely controlled physical tunable delays. Another modulator, biased at its null point and driven by electrical DAC's fourth differential signal, served as MLO. Given that experimental platform is a discrete-component prototype, preliminary manual calibrations including polarization alignment, imperfection pre-emphasis, and electrical/optical delay synchronization were performed. Notably, while high-speed integrated coherent receiver (ICR) and digital storage oscilloscope (DSO) in Fig. 3(a) were needed for prototype's preliminary calibrations, they aren't necessary for the integrated architecture or main skew calibration process.

The sequences for both LSB and MSB were binary pseudo-random sequence and

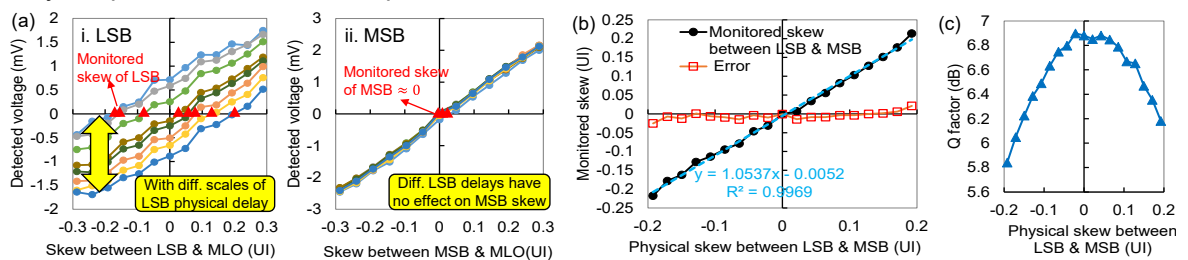


Fig. 5: (a) With different scales of LSB's physical delay, the measured S curves of LSB and MSB are shown in i and ii, respectively. (b) Physically added inter-bit skew versus monitored skew (c) Q factors under different inter-bit skews.

uncorrelated with each other. The MLO sequence was the product of a square wave $sq[n]$ and the designed $B[n]$. As in Fig. 3(b), one frame (128k symbols) of the MLO sequence was partitioned into four distinct segments, designed for LSB phase estimation, LSB skew monitoring, MSB phase estimation, and MSB skew monitoring, respectively.

After performing the steps outlined in Tab. 1, the detected voltage V_0^p was obtained, where p denotes the bit. With various digitally added skews, V_0^p was proportional to the skew τ_p between BUT and MLO, as depicted in Fig. 4(a), enabling indication of both magnitude and direction. The slope of the "S" curve for MSB is approximately twice that of LSB, reflecting their respective bit weights. Furthermore, comprehensive measurements of the entire "S" curves, as shown in Fig. 4(b), demonstrate that the effective range of the skew calibration covers at least ± 0.5 UI.

To avoid circular self-verifications using digital skews, physical delays of the LSB were adjusted to set various skew values. Fig. 5(a) demonstrates that skews between BUT and MLO can be effectively monitored using our proposed method. Fig. 5(a, i) displays the monitored LSB curves, with intersection points at different positions along the horizontal axis. In contrast, the monitored skews of MSB in Fig. 5(a, ii) are nearly zero. Fig. 5(b) presents a comparison between the actual physical inter-bit skews and the monitored ones; a fitting slope of 1.05 and maximum error of ± 0.025 UI validate the accuracy of the skew calibration. As illustrated in Fig. 5(c), this degree of inter-bit skew has negligible impact on Q performance, supporting stable operation of the optical DAC transmitter.

Conclusions

A power-efficient in-service skew calibration, based on an MLO and low-speed coherent detection, is presented for the transmitters with segmented modulator. Experimental validation on a PAM-4 Si optical DAC transmitter shows that proposed method accurately provides skew magnitude and sign across a ± 0.5 UI range. Within ± 0.2 UI, monitored skew agrees well with physical delay, exhibiting a maximum error of ± 0.025 UI.

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